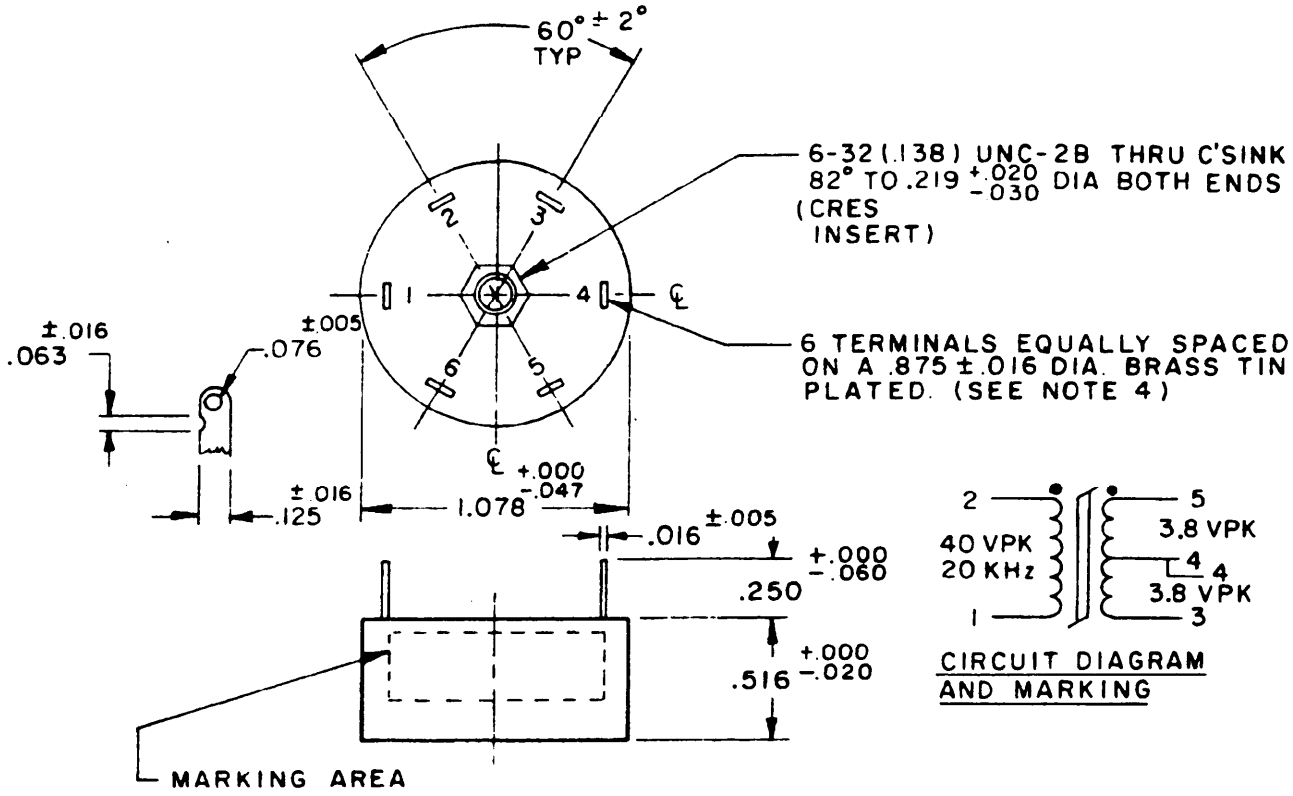
MILITARY SPECIFICATION SHEET

TRANSFORMER, SATURABLE, TF5S40ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on the side of case.
4. Brass, 70% Cu, 30% Zn, per Federal Specification QQ-B-613.

INCHES	MM
.005	.13
.016	.41
.020	.51
.030	.76
.047	1.19
.060	1.52
.063	1.60
.076	1.93
.125	3.18
.138	3.51
.219	5.56
.250	6.35
.516	13.11
.875	22.23
1.078	27.38

FIGURE 1. Dimensions and configuration.

FSC 5950

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-2): 40 volts peak, 20 kilohertz square wave.

~~Volt-ampere rating: 24 volt-amperes (no load).~~

Secondary voltage (3-5): 7.6 volts peak ct.

Secondary current (3-5): 1 ampere average.

Working voltage (peak): (1-2): 40 volts.
(3-5): 7.6 volts.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Terminals: Solder terminal.

Terminal height: $0.250\ ^{+0.000}_{-0.060}$ inch.

Weight: 28.4 grams, maximum.

Operating temperature range: -55° to +130°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2.0 pounds.

Dielectric withstanding voltage (at sea level): 100 volts dc.

Electrical characteristics:

No load: With 40 volts peak, 20 kilohertz square wave across (1-2), current in (1-2) shall not exceed 750 milliamperes and the power in (1-2) shall not exceed 30 watts.

Voltage across (3-5): 7.8 volts ct ±3 percent.

Rated load: With 40 volts, 20 kilohertz square wave across (1-2), the voltage across (3-5) shall be 7.6 volts ct ±5 percent.

DC resistance: (1-2) .501 to .679 ohms max.
(3-4) .0116 to .0156 ohms max.
(4-5) .0116 to .0156 ohms max.

Polarity: Additive with 2 and 3 connected.

Temperature rise: 50°C with 40 volts peak, 20 kilohertz square wave across (1-2) at an ambient temperature of 80°C.

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Vibration (high frequency): MIL-STD-202, method 204, test condition A.

Shock (specified pulse): MIL-STD-202, method 213, test condition H.

Marking location: See figure 1.

Part number: M27/309-01.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-T-27/309 TRANSFORMER, SATURABLE, TF5S4077

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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